

# Summarizing Data

## Computational Mathematics and Statistics

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# One Minute Paper Results

What was the most important thing you learned during this class?

What important question remains unanswered for you?

# Data Types / Descriptives / Visualizations

| Data Type                  | Descriptive Stats                             | Visualization                |
|----------------------------|-----------------------------------------------|------------------------------|
| Continuous                 | mean, median, mode, standard deviation, IQR   | histogram, density, box plot |
| Discrete                   | contingency table, proportional table, median | bar plot                     |
| Categorical                | contingency table, proportional table         | bar plot                     |
| Ordinal                    | contingency table, proportional table, median | bar plot                     |
| Two quantitative           | correlation                                   | scatter plot                 |
| Two qualitative            | contingency table, chi-squared                | mosaic plot, bar plot        |
| Quantitative & Qualitative | grouped summaries, ANOVA, t-test              | box plot                     |

# Variance

Population Variance:

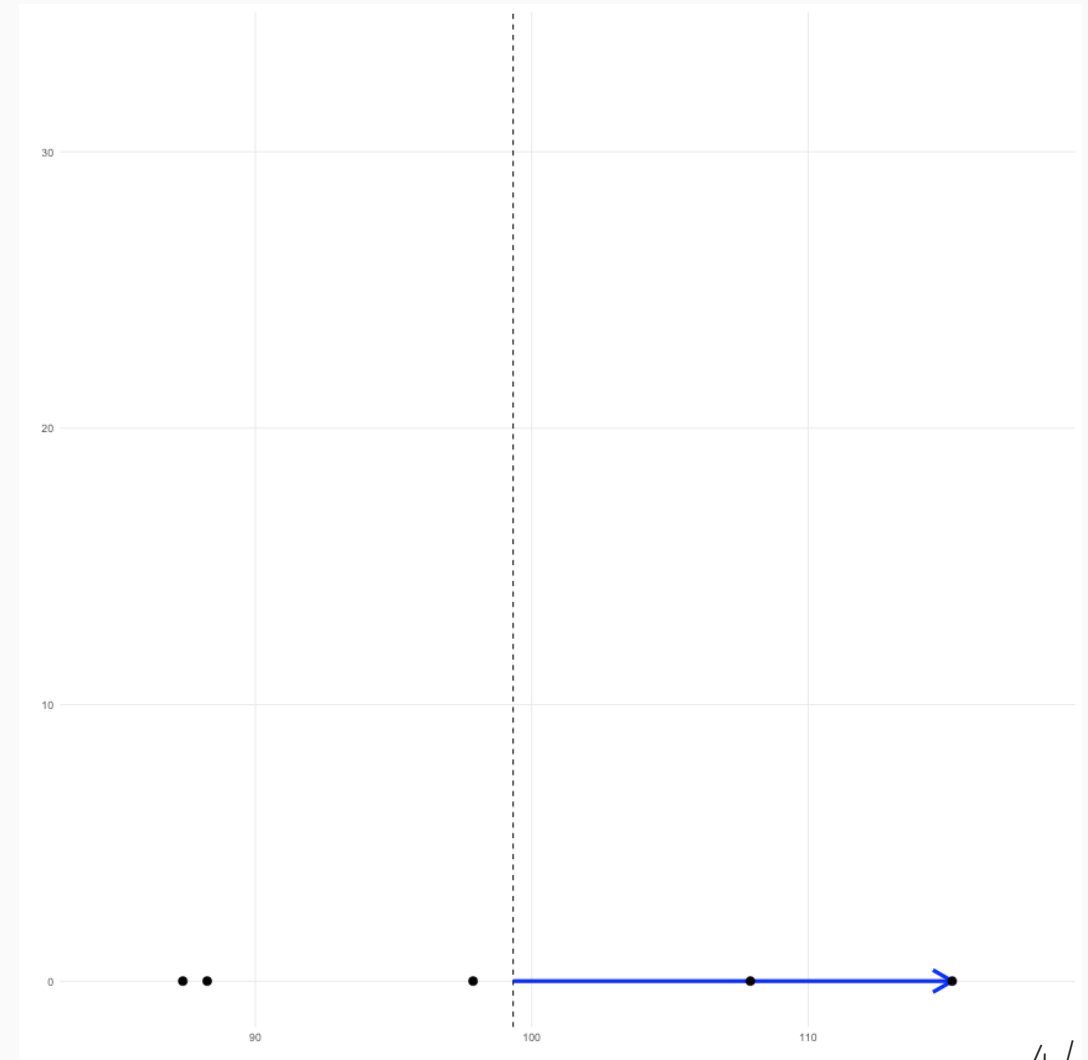
$$S^2 = \frac{\sum (x_i - \bar{x})^2}{N}$$

Consider a dataset with five values (black points in the figure). For the largest value, the deviance is represented by the blue line (  $x_i - \bar{x}$  ).

See also:

<https://shiny.rit.albany.edu/stat/visualizess/>

<https://github.com/jbryer/VisualStats/>

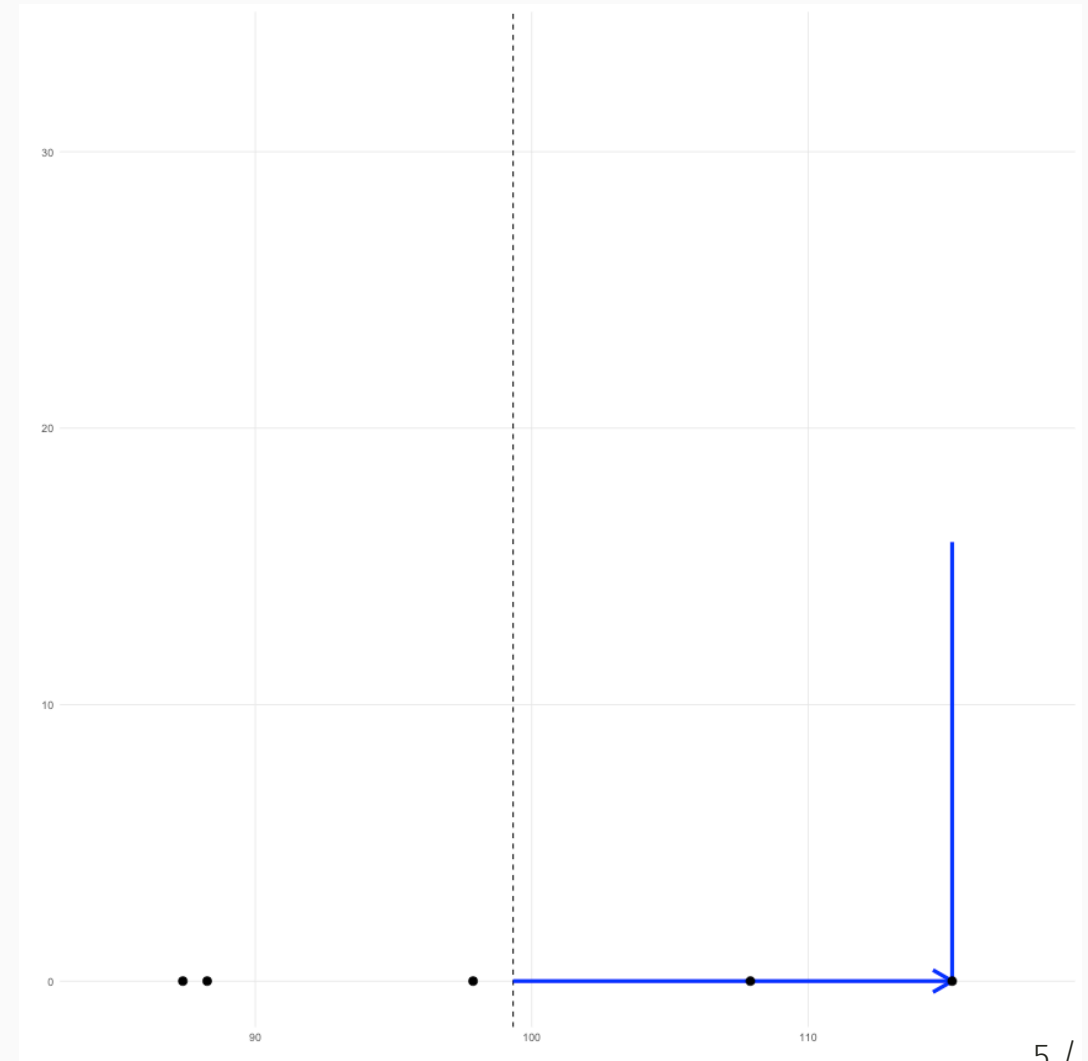


# Variance (cont.)

Population Variance:

$$S^2 = \frac{\sum (x_i - \bar{x})^2}{N}$$

In the numerator, we square each of these deviances. We can conceptualize this as a square. Here, we add the deviance in the y direction.

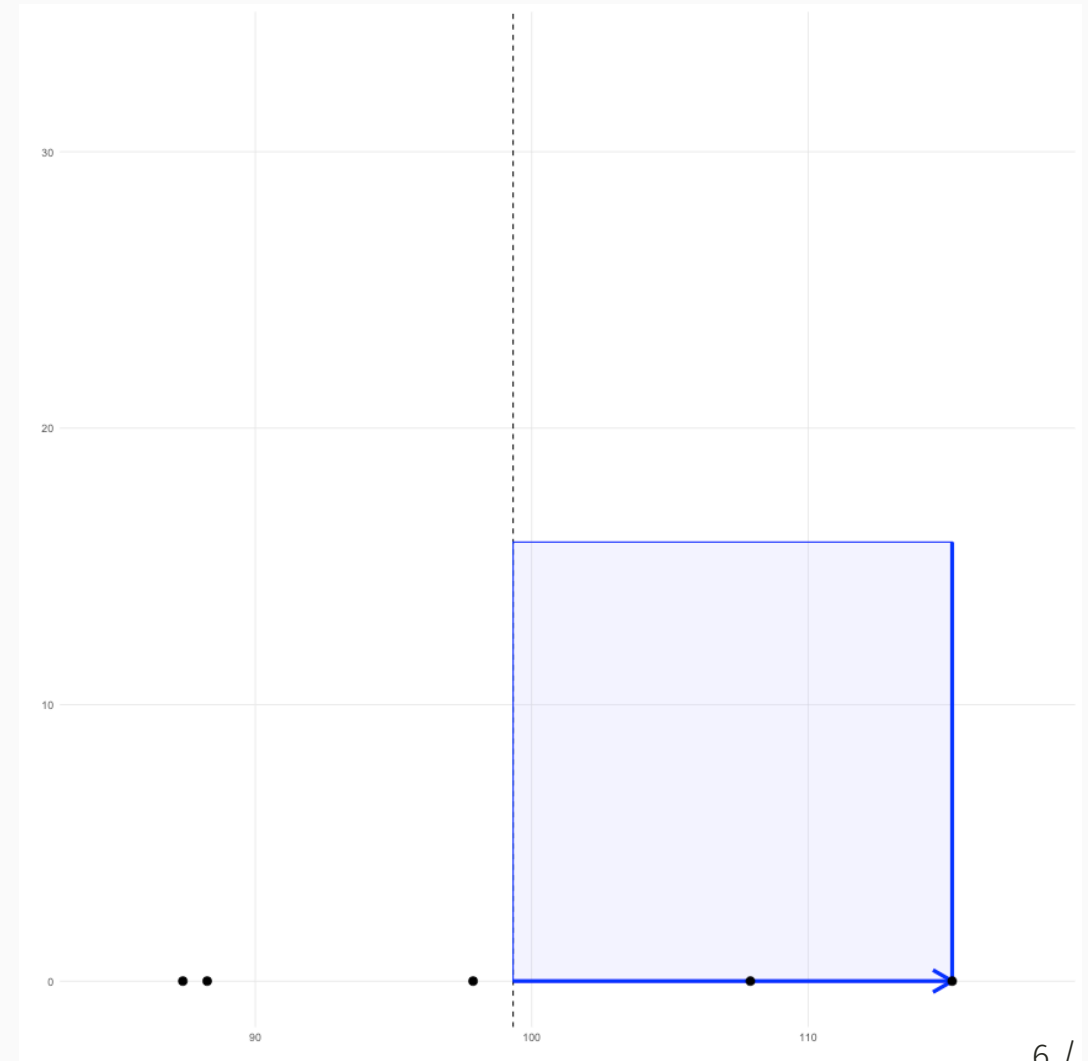


# Variance (cont.)

Population Variance:

$$S^2 = \frac{\sum (x_i - \bar{x})^2}{N}$$

We end up with a square.

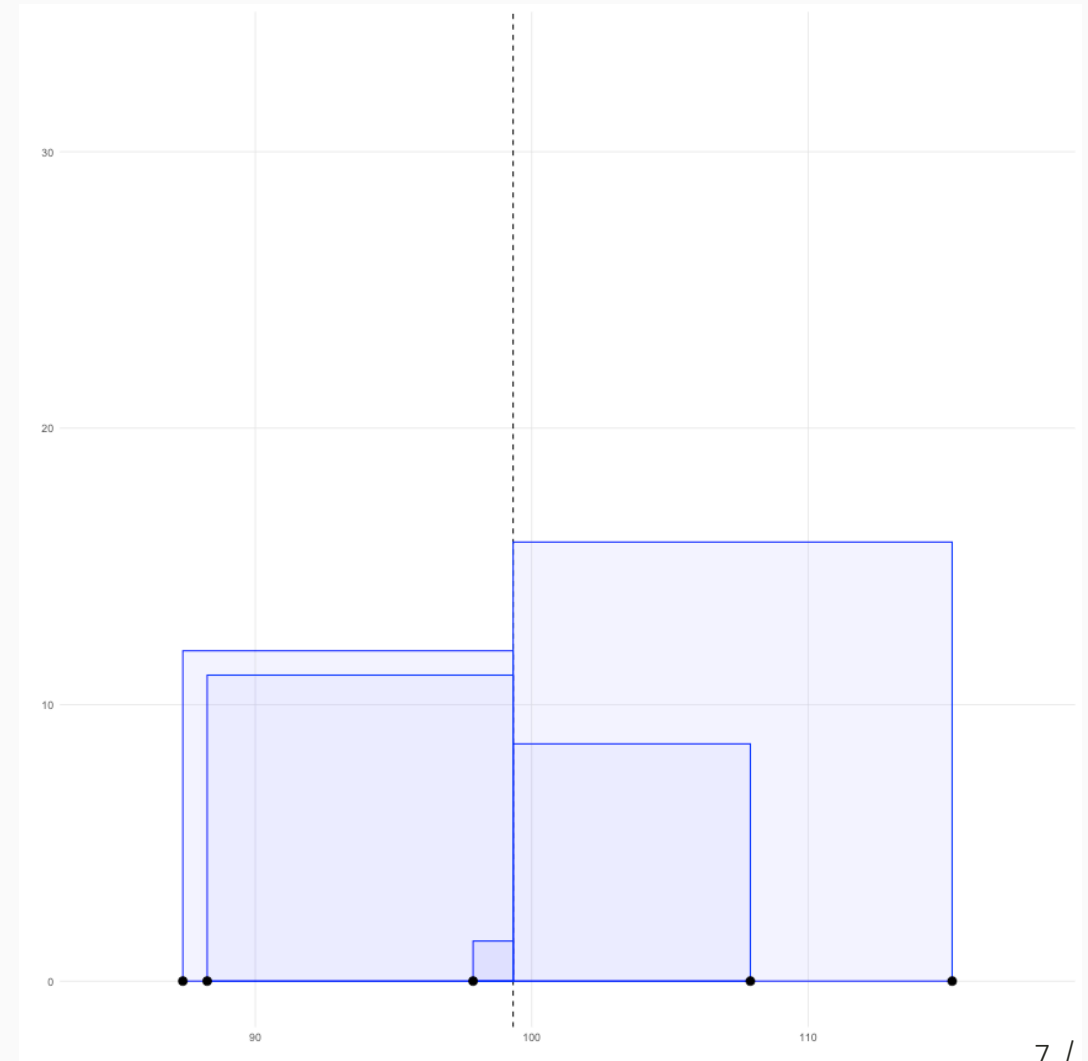


# Variance (cont.)

Population Variance:

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{N}$$

We can plot the squared deviance for all the data points. That is, each component in the numerator is the area of each of these squares.

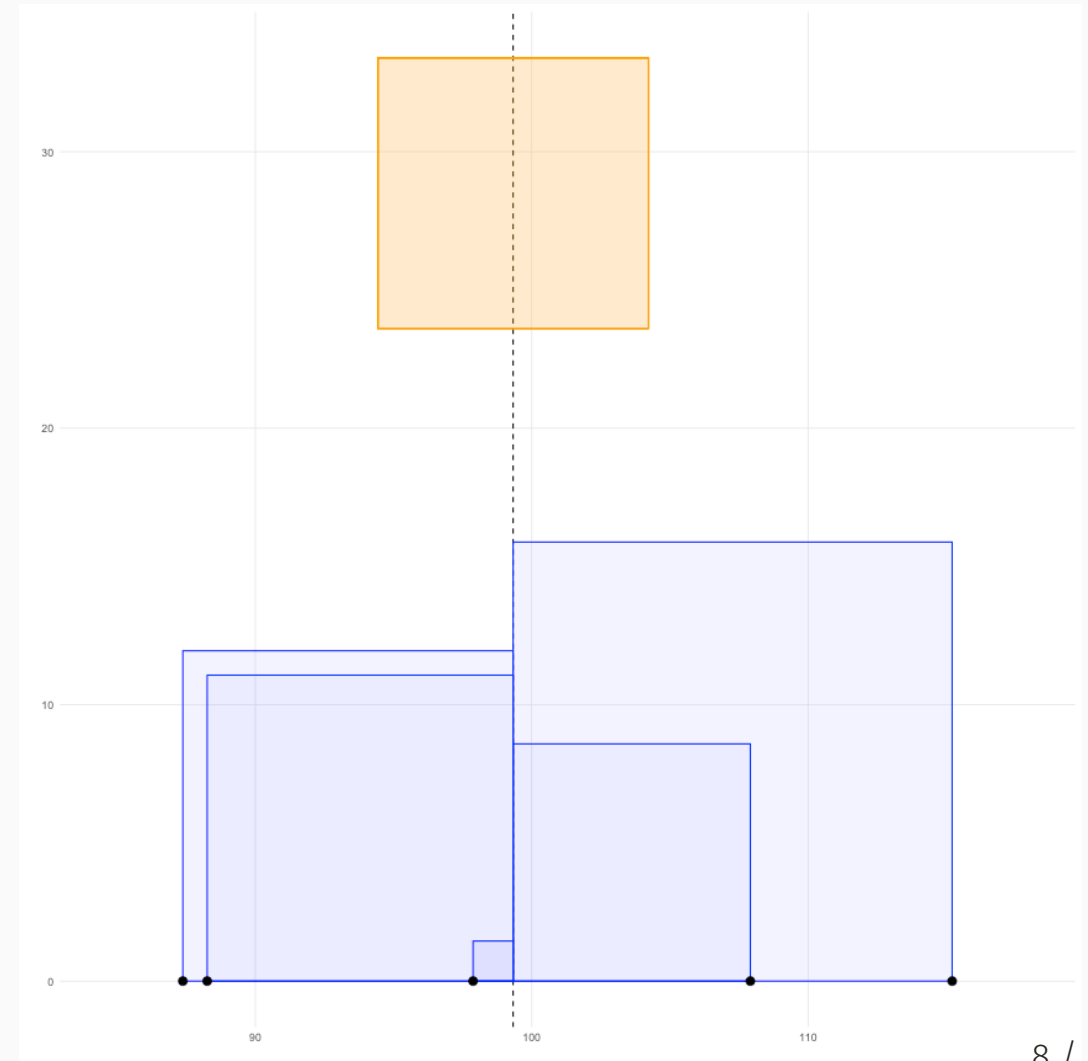


# Variance (cont.)

Population Variance:

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{N}$$

The variance is therefore the average of the area of all these squares, here represented by the orange square.





# Population versus Sample Variance

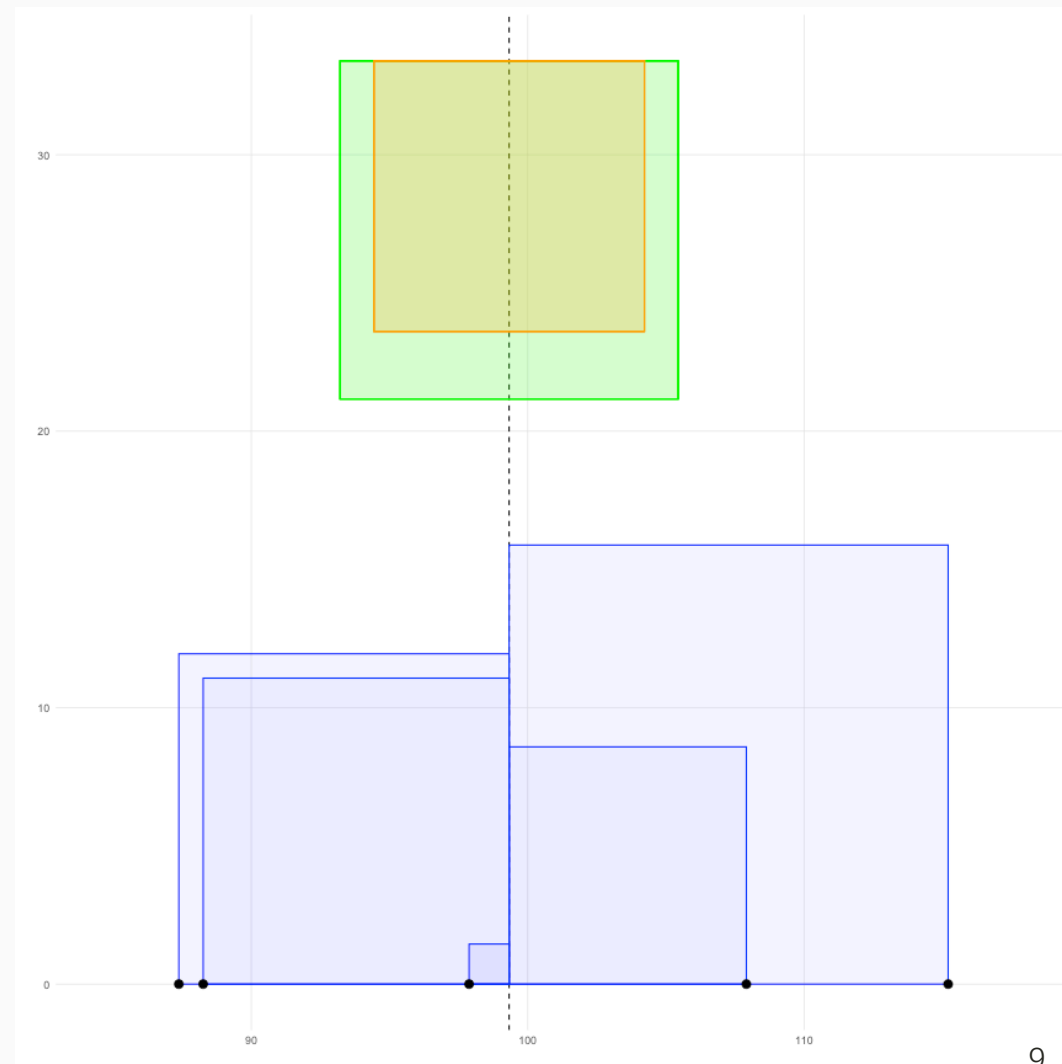
Typically we want the sample variance. The difference is we divide by  $n - 1$  to calculate the sample variance. This results in a slightly larger area (variance) then if we divide by  $n$ .

Population Variance (yellow):

$$S^2 = \frac{\Sigma(x_i - \bar{x})^2}{N}$$

Sample Variance (green):

$$s^2 = \frac{\Sigma(x_i - \bar{x})^2}{n - 1}$$



# Robust Statistics

Consider the following data randomly selected from the normal distribution:

```
set.seed(41)
x <- rnorm(30, mean = 100, sd = 15)
mean(x); sd(x)
```

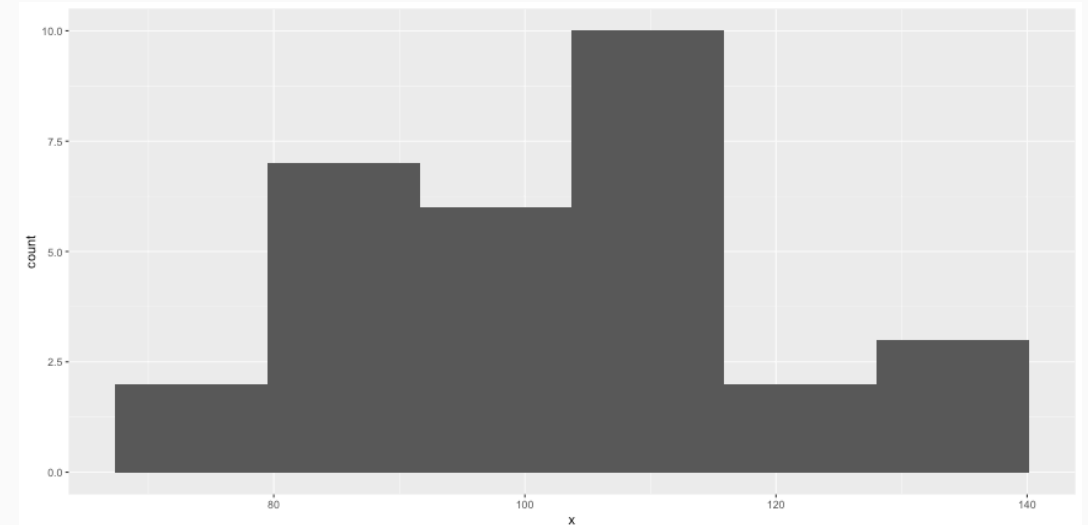
```
## [1] 103.1934
```

```
## [1] 16.8945
```

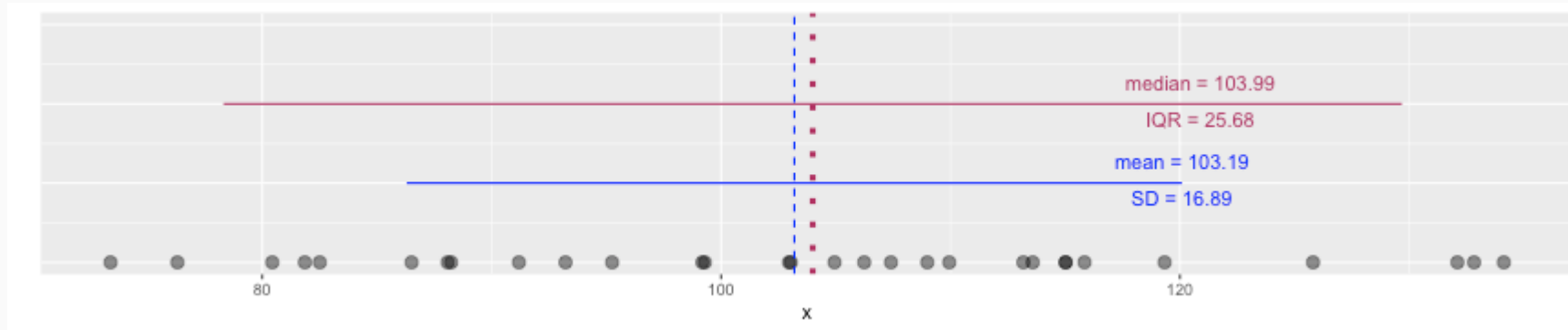
```
median(x); IQR(x)
```

```
## [1] 103.9947
```

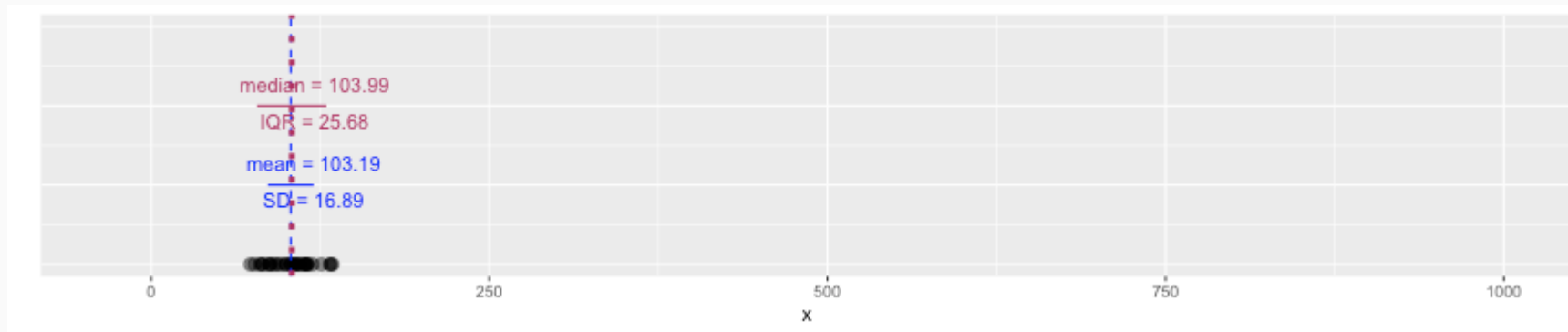
```
## [1] 25.68004
```



# Robust Statistics

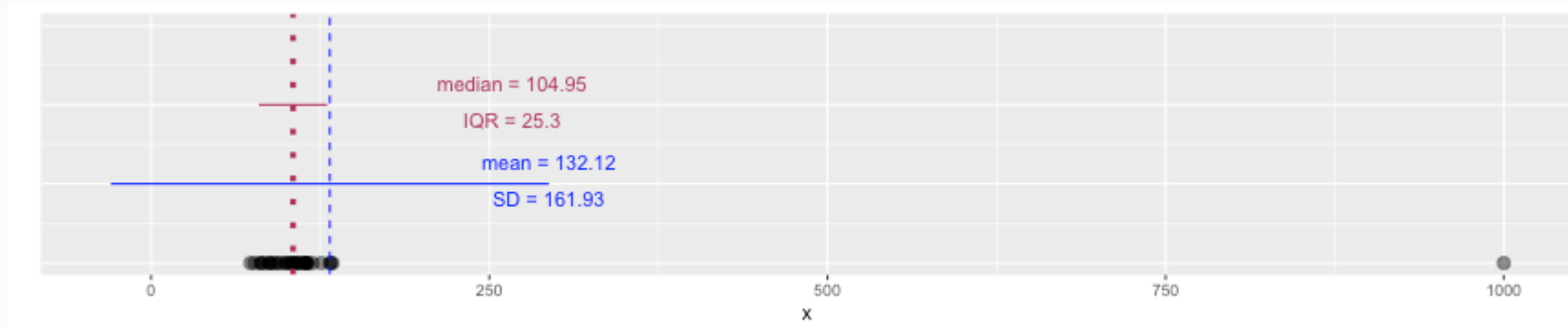


# Robust Statistics



Let's add an extreme value:

```
x <- c(x, 1000)
```



# Robust Statistics

Median and IQR are more robust to skewness and outliers than mean and SD. Therefore,

- for skewed distributions it is often more helpful to use median and IQR to describe the center and spread
- for symmetric distributions it is often more helpful to use the mean and SD to describe the center and spread

# Grammar of Graphics



- `ggplot2` is an R package that provides an alternative framework based upon Wilkinson's (2005) Grammar of Graphics.
- `ggplot2` is, in general, more flexible for creating "prettier" and complex plots.
- Works by creating layers of different types of objects/geometries (i.e. bars, points, lines, polygons, etc.) `ggplot2` has at least three ways of creating plots:
  1. `qplot`
  2. `ggplot(...) + geom_XXX(...) + ...`
  3. `ggplot(...) + layer(...)`
- We will focus only on the second.

# Parts of a ggplot2 Statement



- Data

```
ggplot(myDataFrame, aes(x=x, y=y))
```

- Layers

```
geom_point(), geom_histogram()
```

- Facets

```
facet_wrap(~ cut), facet_grid(~ cut)
```

- Scales

```
scale_y_log10()
```

- Other options

```
ggtitle('my title'), ylim(c(0, 10000)), xlab('x-axis label')
```



# Lots of geoms



```
ls('package:ggplot2')[grep('^geom_', ls('package:ggplot2'))]
```

```
## [1] "geom_abline"      "geom_area"        "geom_bar"         "geom_bin_2d"
## [5] "geom_bin2d"       "geom_blank"       "geom_boxplot"     "geom_col"
## [9] "geom_contour"     "geom_contour_filled" "geom_count"       "geom_crossbar"
## [13] "geom_curve"       "geom_density"     "geom_density_2d"  "geom_density_2d_filled"
## [17] "geom_density2d"   "geom_density2d_filled" "geom_dotplot"     "geom_errorbar"
## [21] "geom_errorbarh"   "geom_freqpoly"    "geom_function"    "geom_hex"
## [25] "geom_histogram"   "geom_hline"       "geom_jitter"      "geom_label"
## [29] "geom_line"        "geom_linerange"   "geom_map"         "geom_path"
## [33] "geom_point"       "geom_pointrange"  "geom_polygon"     "geom_qq"
## [37] "geom_qq_line"     "geom_quantile"    "geom_raster"      "geom_rect"
## [41] "geom_ribbon"      "geom_rug"         "geom_segment"     "geom_sf"
## [45] "geom_sf_label"    "geom_sf_text"     "geom_smooth"      "geom_spoke"
## [49] "geom_step"        "geom_text"        "geom_tile"        "geom_violin"
## [53] "geom_vline"
```

# Data Visualization Cheat Sheet



## Data Visualization with ggplot2 : : CHEAT SHEET



### Basics

ggplot2 is based on the **grammar of graphics**, the idea that you can build every graph from the same components: a **data set**, a **coordinate system**, and **geoms**—visual marks that represent data points.



To display values, map variables in the data to visual properties of the geom (**aesthetics**) like **size**, **color**, and **x** and **y** locations.



Complete the template below to build a graph.

```
ggplot(data = <DATA>) +  
  <GEOM_FUNCTION>(mapping = aes(<MAPPINGS>)) +  
  stat = <STAT>, position = <POSITION> +  
  <COORDINATE_FUNCTION> +  
  <FACET_FUNCTION> +  
  <SCALE_FUNCTION> +  
  <THEME_FUNCTION>
```

ggplot(data = mpg, aes(x = cty, y = hwy)) Begins a plot that you finish by adding layers to. Add one geom function per layer.

ggplot(x = cty, y = hwy, data = mpg, geom = "point") Creates a complete plot with given data, geom, and mappings. Supplies many useful defaults.

last\_plot() Returns the last plot

ggsave("plot.png", width = 5, height = 5) Saves last plot as 5' x 5' file named "plot.png" in working directory. Matches file type to file extension.

### Geoms

Use a geom function to represent data points, use the geom's aesthetic properties to represent variables. Each function returns a layer.

#### GRAPHICAL PRIMITIVES

```
a <- ggplot(economics, aes(date, unemployment))  
b <- ggplot(seals, aes(x = long, y = lat))
```

**a + geom\_blank()**  
(Useful for expanding limits)

**b + geom\_curve()**(aes(yend = lat + 1, xend = long + 1, curvature = z)) - x, yend, y, yend, alpha, angle, color, curvature, linetype, size

**a + geom\_path()**(lineend = "butt", linejoin = "round", linemitre = 1)  
x, y, alpha, color, group, linetype, size

**a + geom\_polygon()**(aes(group = group))  
x, y, alpha, color, fill, group, linetype, size

**b + geom\_rect()**(aes(xmin = long, ymin = lat, xmax = long + 1, ymax = lat + 1)) - xmax, xmin, ymax, ymin, alpha, color, fill, linetype, size

**a + geom\_ribbon()**(aes(ymin = unemployment - 900, ymax = unemployment + 900)) - x, ymax, ymin, alpha, color, fill, group, linetype, size

#### LINE SEGMENTS

common aesthetics: x, y, alpha, color, linetype, size

```
b + geom_abline(aes(intercept = 0, slope = 1))  
b + geom_hline(aes(yintercept = lat))  
b + geom_vline(aes(xintercept = long))
```

```
b + geom_segment(aes(yend = lat + 1, xend = long + 1))  
b + geom_spoke(aes(angle = 1:1155, radius = 1))
```

#### ONE VARIABLE continuous

```
c <- ggplot(mpg, aes(hwy)); c2 <- ggplot(mpg)
```

**c + geom\_area()**(stat = "bin")  
x, y, alpha, color, fill, linetype, size

**c + geom\_density()**(kernel = "gaussian")  
x, y, alpha, color, fill, group, linetype, size, weight

**c + geom\_dotplot()**  
x, y, alpha, color, fill

**c + geom\_freqpoly()** x, y, alpha, color, group, linetype, size

**c + geom\_histogram()**(binwidth = 5) x, y, alpha, color, fill, linetype, size, weight

**c2 + geom\_qq()**(aes(sample = hwy)) x, y, alpha, color, fill, linetype, size, weight

#### discrete

```
d <- ggplot(mpg, aes(fill))
```

**d + geom\_bar()**  
x, alpha, color, fill, linetype, size, weight

#### TWO VARIABLES

**continuous x, continuous y**  
e <- ggplot(mpg, aes(cty, hwy))

**e + geom\_label()**(aes(label = cty), nudge\_x = 1, nudge\_y = 1, check\_overlap = TRUE) x, y, label, alpha, angle, color, family, fontface, hjust, lineheight, size, vjust

**e + geom\_jitter()**(height = 2, width = 2) x, y, alpha, color, fill, shape, size

**e + geom\_point()**, x, y, alpha, color, fill, shape, size, stroke

**e + geom\_quantile()**, x, y, alpha, color, group, linetype, size, weight

**e + geom\_rug()**(sides = "bl"), x, y, alpha, color, linetype, size

**e + geom\_smooth()**(method = lm), x, y, alpha, color, fill, group, linetype, size, weight

**e + geom\_text()**(aes(label = cty), nudge\_x = 1, nudge\_y = 1, check\_overlap = TRUE) x, y, label, alpha, angle, color, family, fontface, hjust, lineheight, size, vjust

#### discrete x, continuous y

```
f <- ggplot(mpg, aes(class, hwy))
```

**f + geom\_col()**, x, y, alpha, color, fill, group, linetype, size

**f + geom\_boxplot()**, x, y, lower, middle, upper, ymax, ymin, alpha, color, fill, group, linetype, shape, size, weight

**f + geom\_dotplot()**(binaxis = "y", stackdir = "center"), x, y, alpha, color, fill, group

**f + geom\_violin()**(scale = "area"), x, y, alpha, color, fill, group, linetype, size, weight

#### discrete x, discrete y

```
g <- ggplot(diamonds, aes(cut, color))
```

**g + geom\_count()**, x, y, alpha, color, fill, shape, size, stroke

#### THREE VARIABLES

```
seals$z <- with(seals, sqrt(delta_long^2 + delta_lat^2)) l <- ggplot(seals, aes(long, lat))
```

**l + geom\_contour()**(aes(z = z))  
x, y, z, alpha, colour, group, linetype, size, weight

**l + geom\_raster()**(aes(fill = z), hjust = 0.5, vjust = 0.5, interpolate = FALSE)  
x, y, alpha, fill

**l + geom\_tile()**(aes(fill = z)), x, y, alpha, color, fill, linetype, size, width

#### continuous bivariate distribution

```
h <- ggplot(diamonds, aes(carat, price))
```

**h + geom\_bin2d()**(binwidth = c(0.25, 500))  
x, y, alpha, color, fill, linetype, size, weight

**h + geom\_density2d()**  
x, y, alpha, colour, group, linetype, size

**h + geom\_hex()**  
x, y, alpha, colour, fill, size

#### continuous function

```
i <- ggplot(economics, aes(date, unemployment))
```

**i + geom\_area()**  
x, y, alpha, color, fill, linetype, size

**i + geom\_line()**  
x, y, alpha, color, group, linetype, size

**i + geom\_step()**(direction = "hv")  
x, y, alpha, color, group, linetype, size

#### visualizing error

```
df <- data.frame(grp = c("A", "B"), fit = 4.5, se = 1.2)  
j <- ggplot(df, aes(grp, fit, ymin = fit-se, ymax = fit+se))
```

**j + geom\_crossbar()**(fatten = 2)  
x, y, ymax, ymin, alpha, color, fill, group, linetype, size

**j + geom\_errorbar()**, x, ymax, ymin, alpha, color, group, linetype, size, width (also **geom\_errorbarh()**)

**j + geom\_linerange()**  
x, ymin, ymax, alpha, color, group, linetype, size

**j + geom\_pointrange()**  
x, y, ymin, ymax, alpha, color, fill, group, linetype, shape, size

#### maps

```
data <- data.frame(murder = USArrests$Murder,  
state = tolower(rownames(USArrests)))  
map <- map_data("state")  
k <- ggplot(data, aes(fill = murder))
```

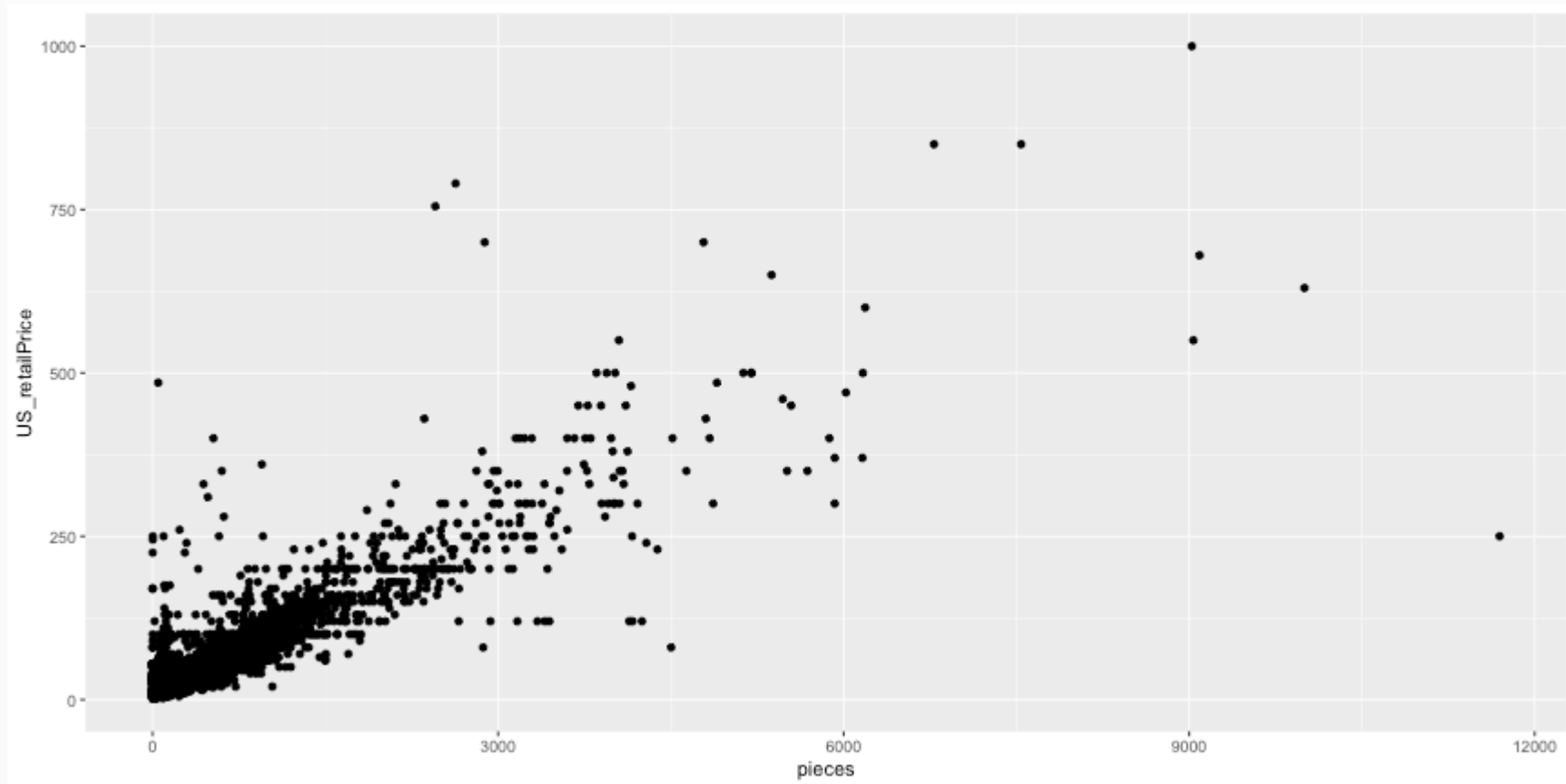
**k + geom\_map()**(aes(map\_id = state), map = map)  
**expand\_limits**(x = map\$long, y = map\$lat),  
map\_id, alpha, color, fill, linetype, size



# Scatterplot



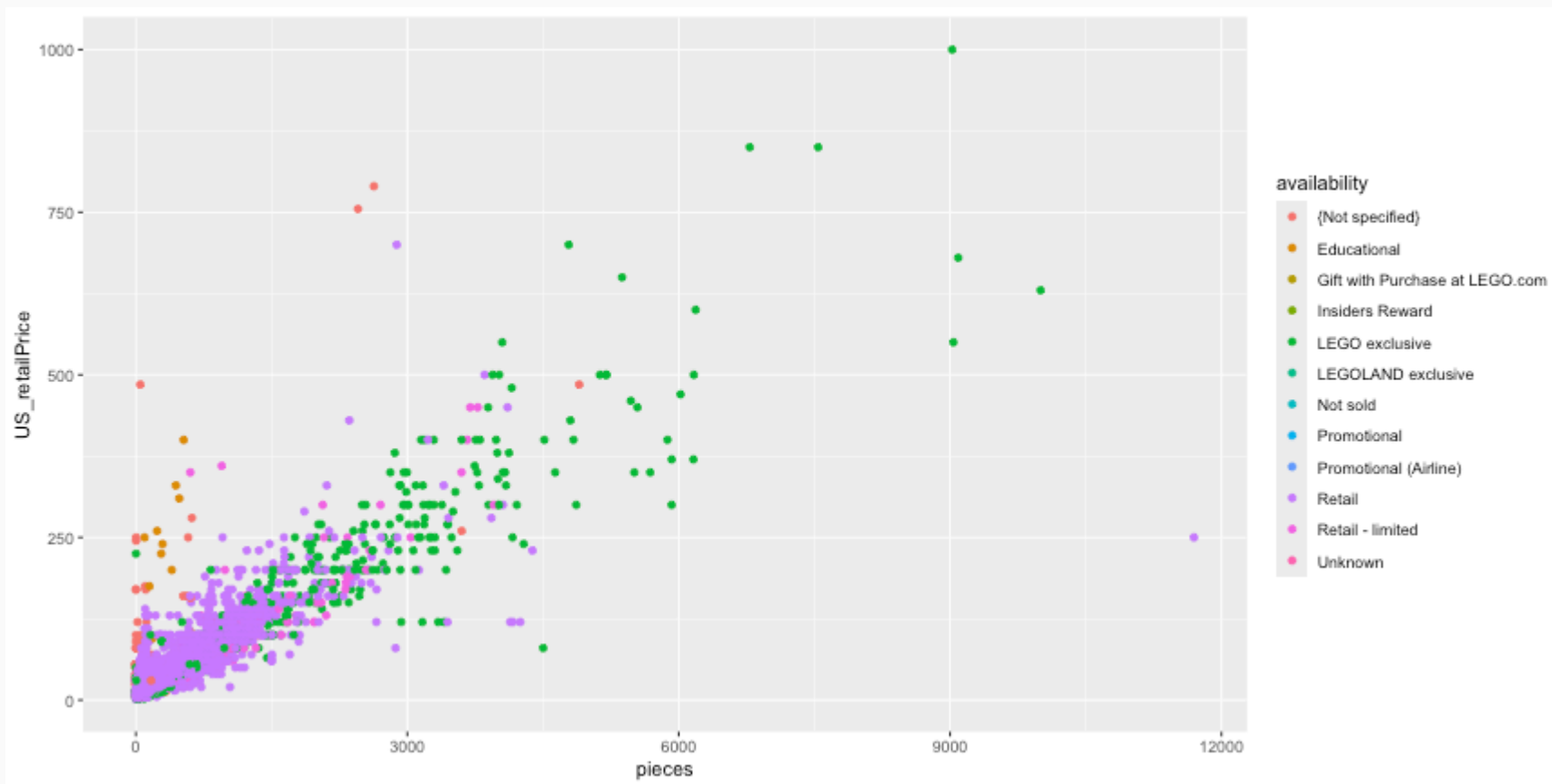
```
ggplot(legosets, aes(x=pieces, y=US_retailPrice)) + geom_point()
```



# Scatterplot (cont.)



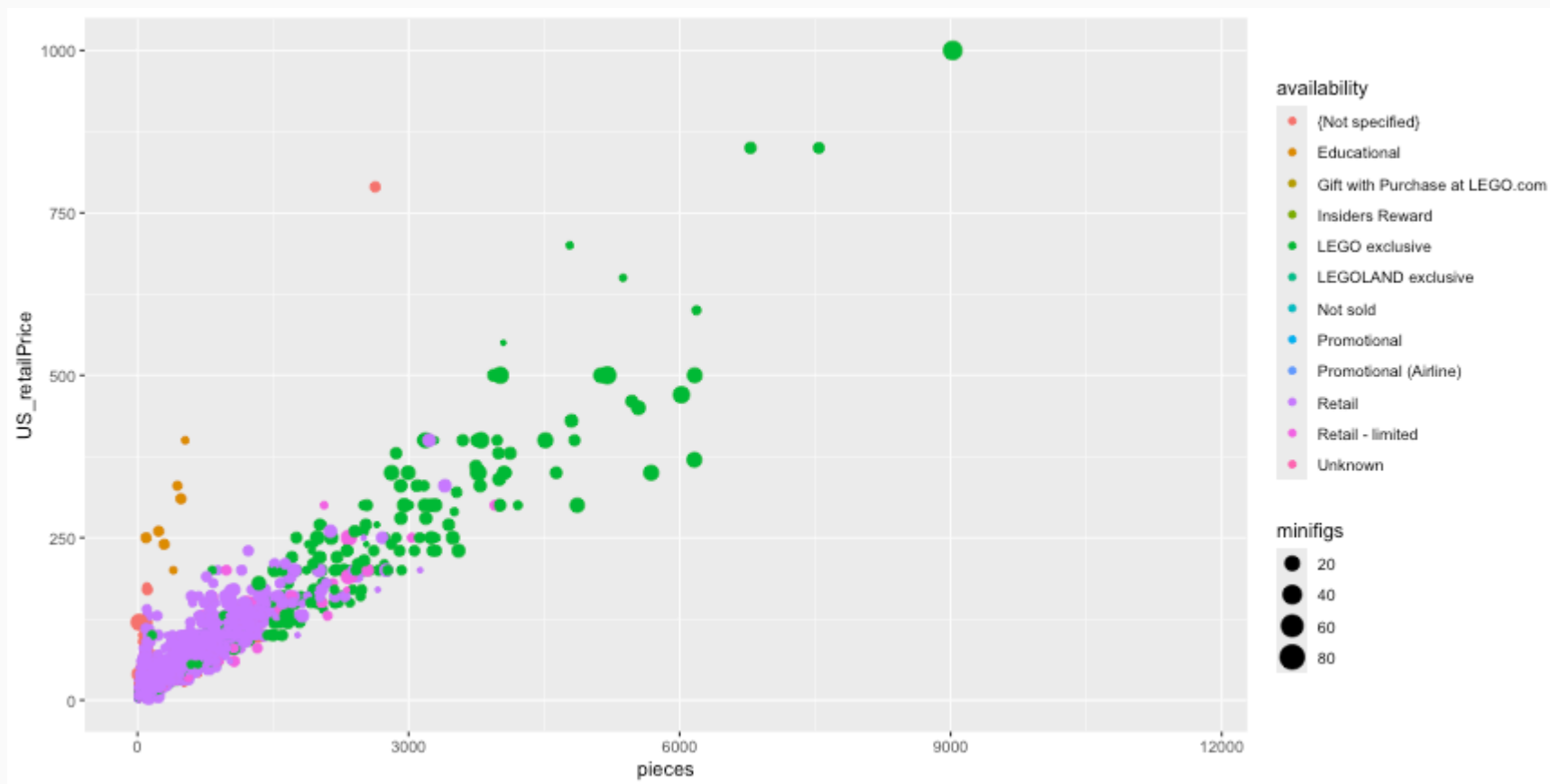
```
ggplot(legosets, aes(x=pieces, y=US_retailPrice, color=availability)) + geom_point()
```



# Scatterplot (cont.)



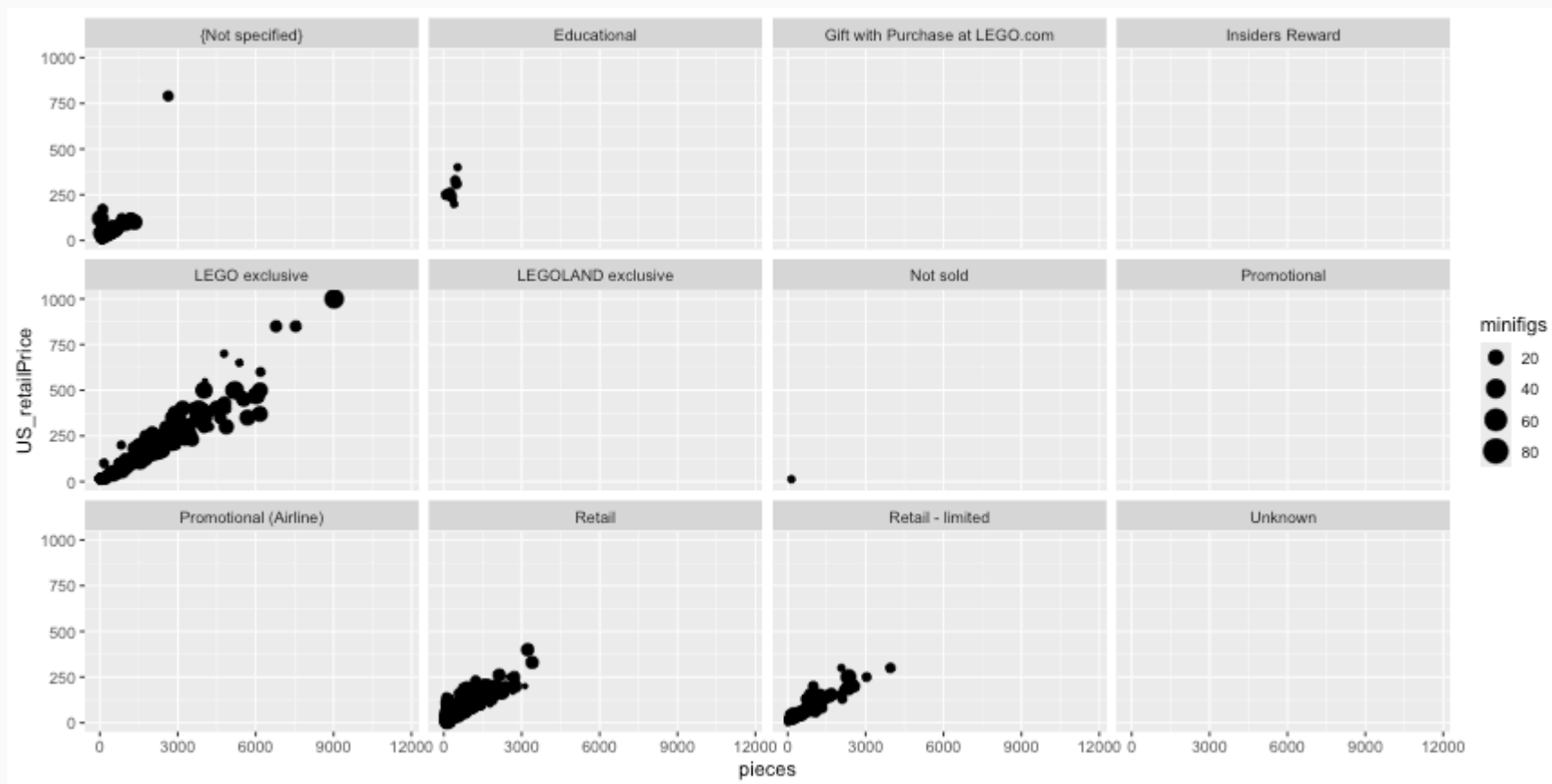
```
ggplot(legosets, aes(x=pieces, y=US_retailPrice, size=minifigs, color=availability)) + geom_point()
```



# Scatterplot (cont.)



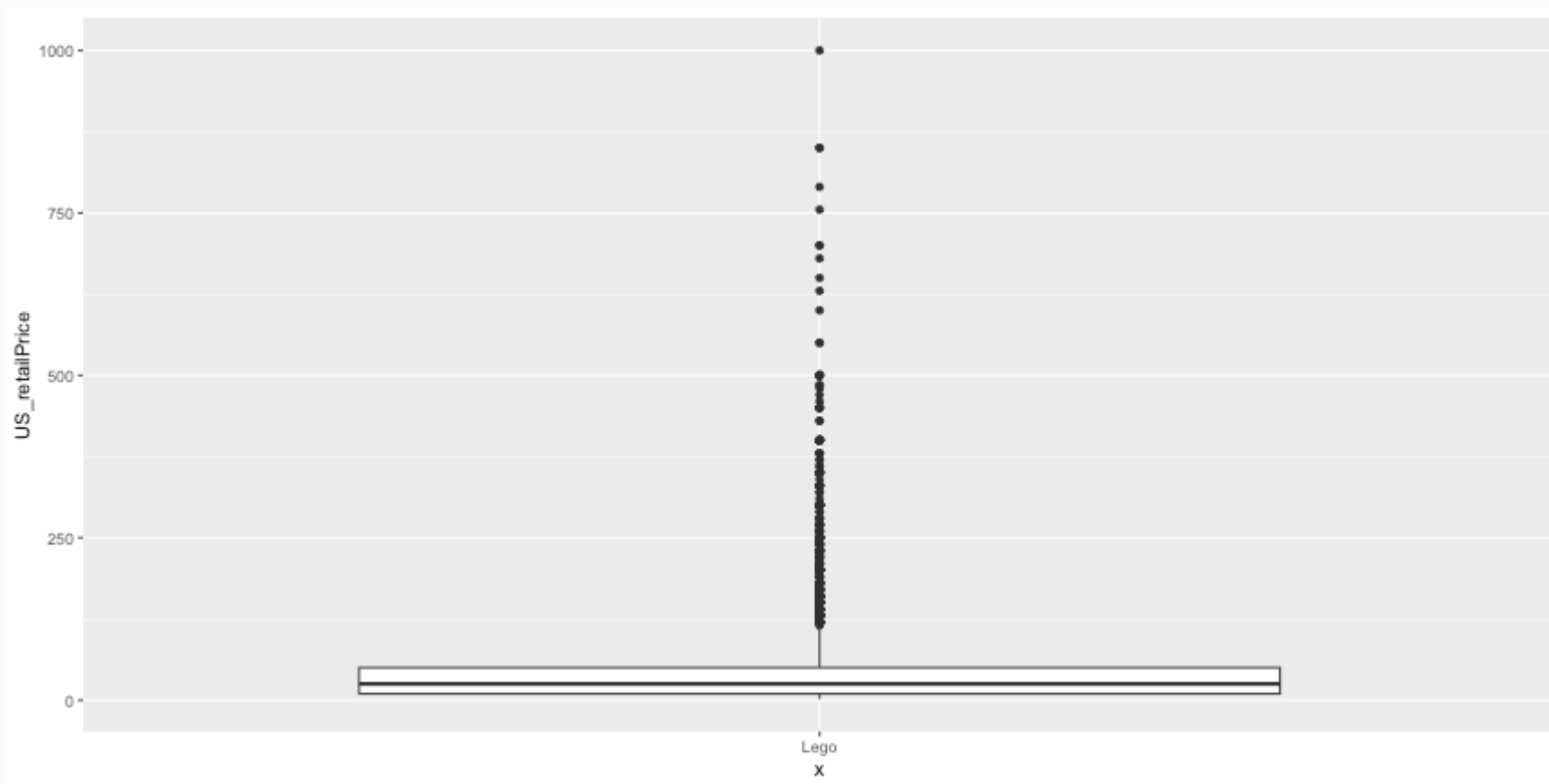
```
ggplot(legosets, aes(x=pieces, y=US_retailPrice, size=minifigs)) + geom_point() + facet_wrap(~ availability)
```



# Boxplots



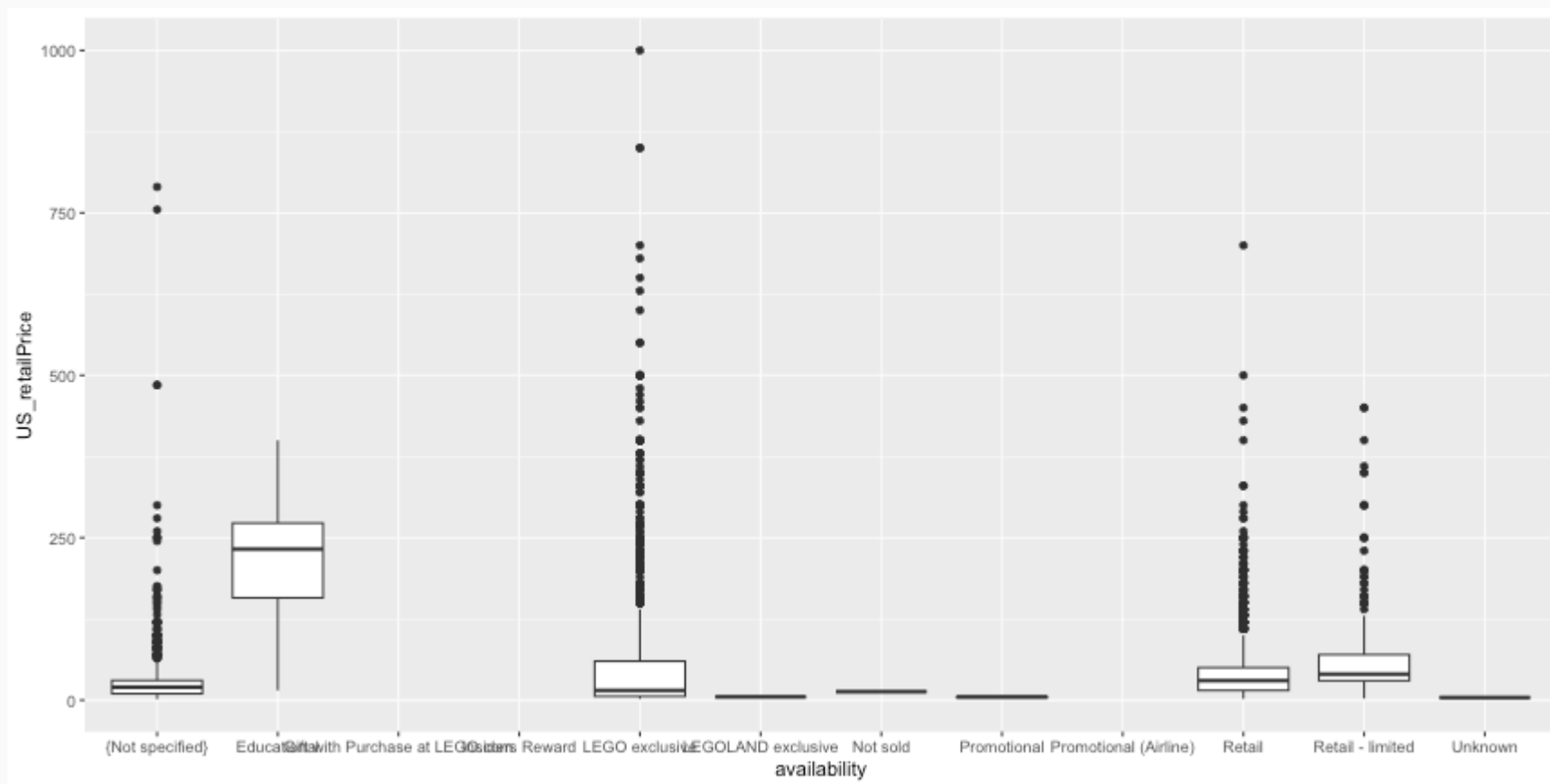
```
ggplot(legosets, aes(x='Lego', y=US_retailPrice)) + geom_boxplot()
```



# Boxplots (cont.)



```
ggplot(legosets, aes(x=availability, y=US_retailPrice)) + geom_boxplot()
```

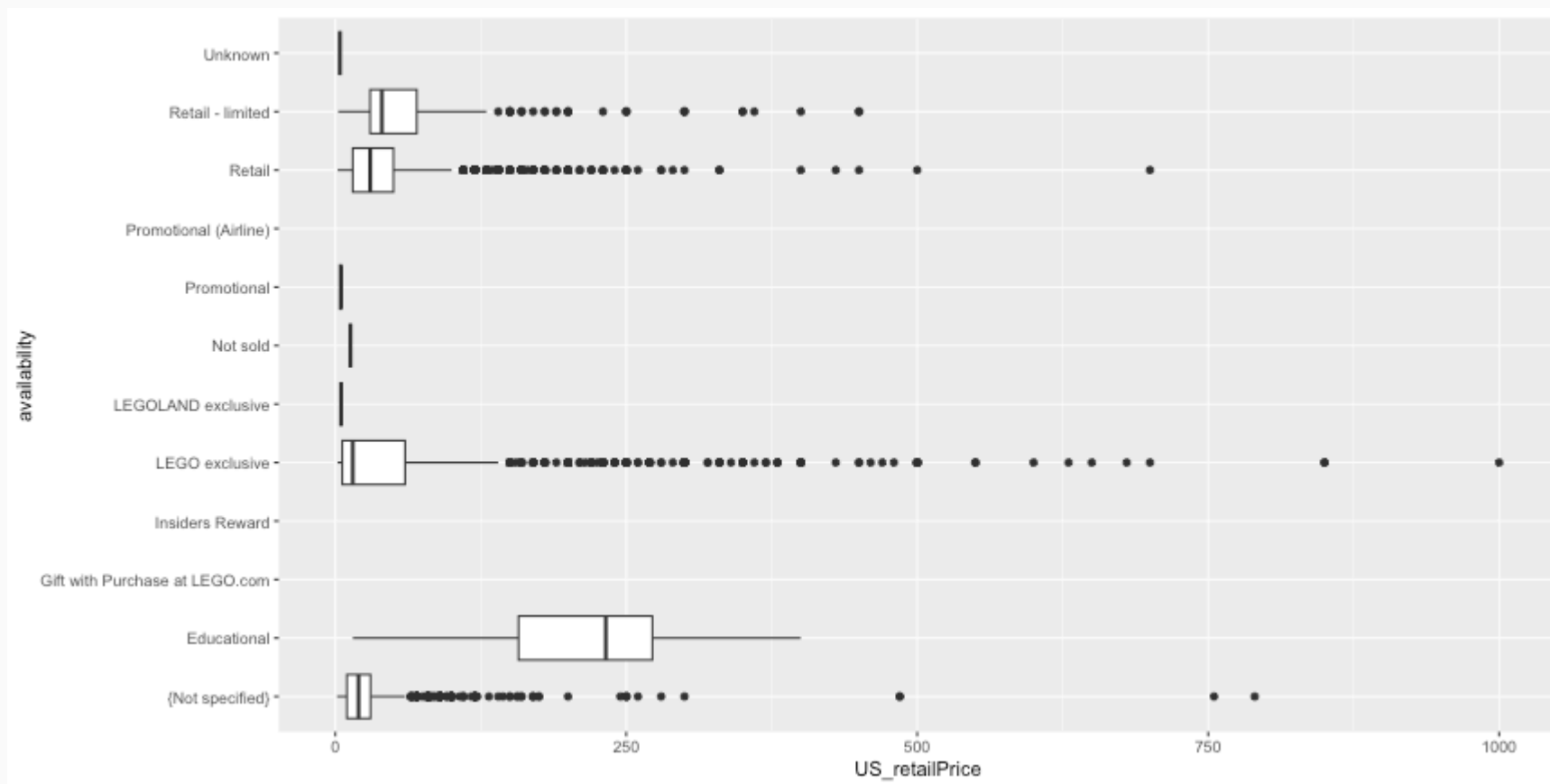




# Boxplot (cont.)



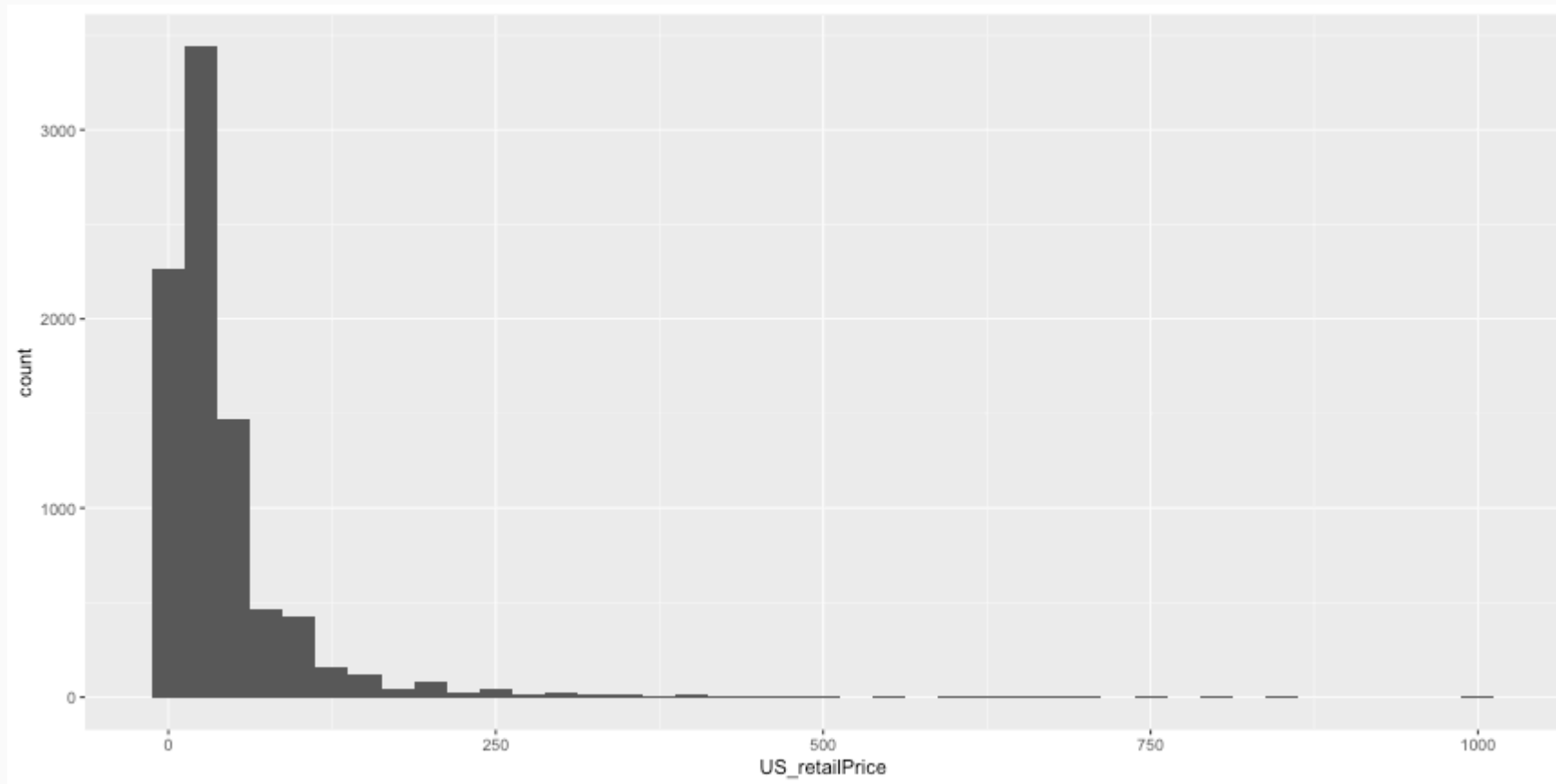
```
ggplot(legosets, aes(x=availability, y=US_retailPrice)) + geom_boxplot() + coord_flip()
```



# Histograms



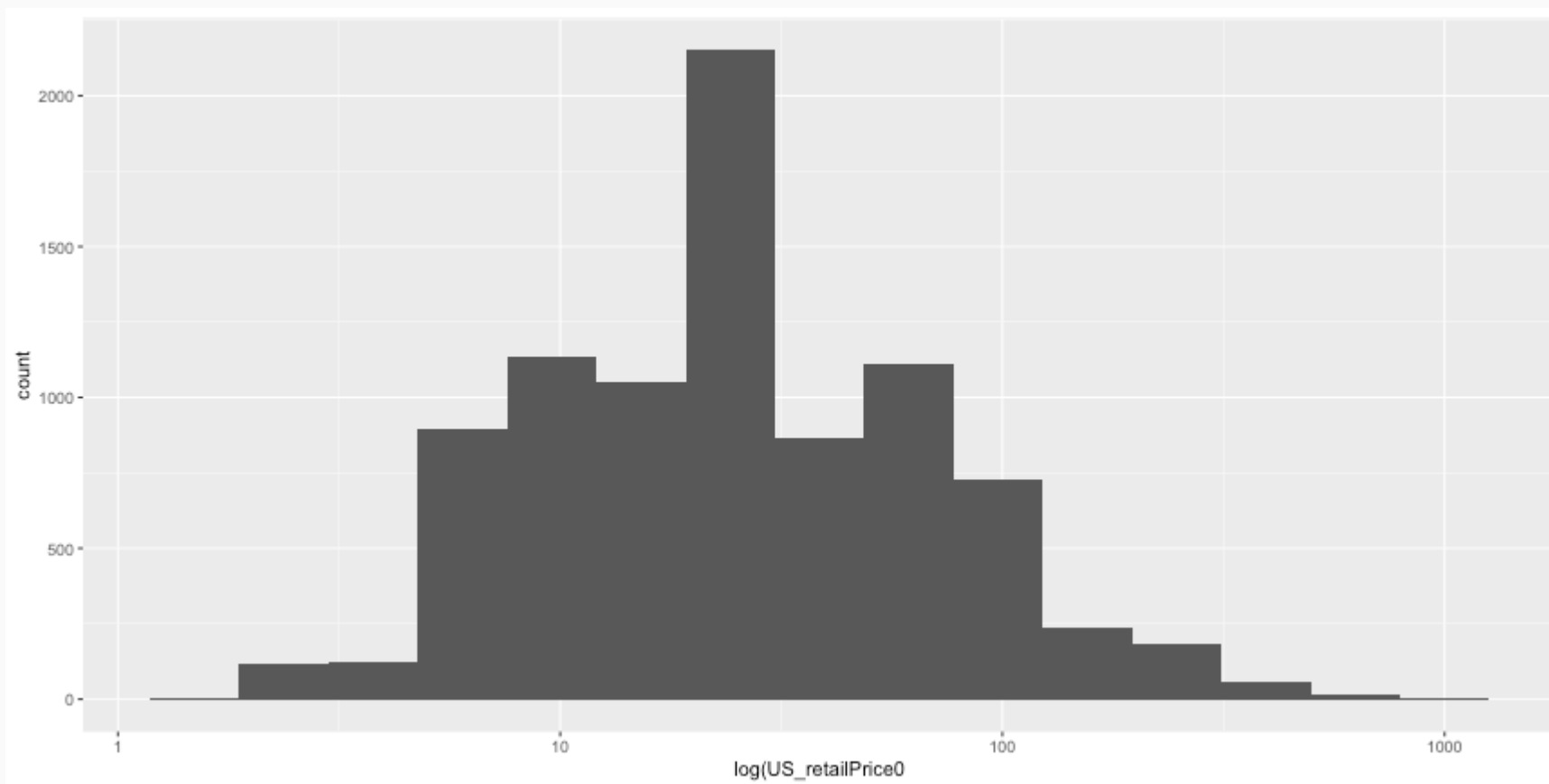
```
ggplot(legosets, aes(x = US_retailPrice)) + geom_histogram(binwidth = 25)
```



# Histograms (cont.)



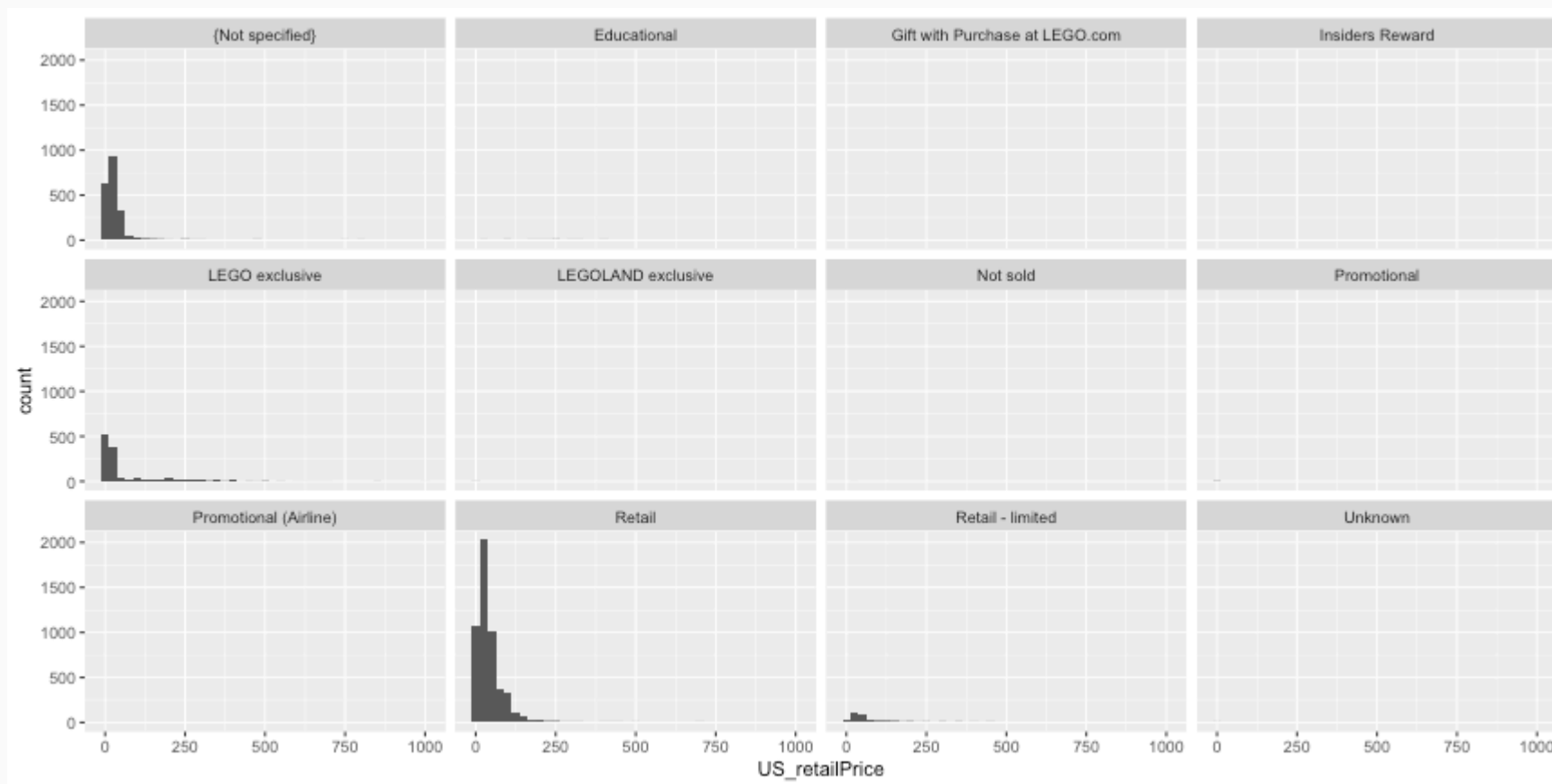
```
ggplot(legosets, aes(x = US_retailPrice)) + geom_histogram(bins = 15) + scale_x_log10() + xlab('log(US_retailPrice0)')
```



# Histograms (cont.)



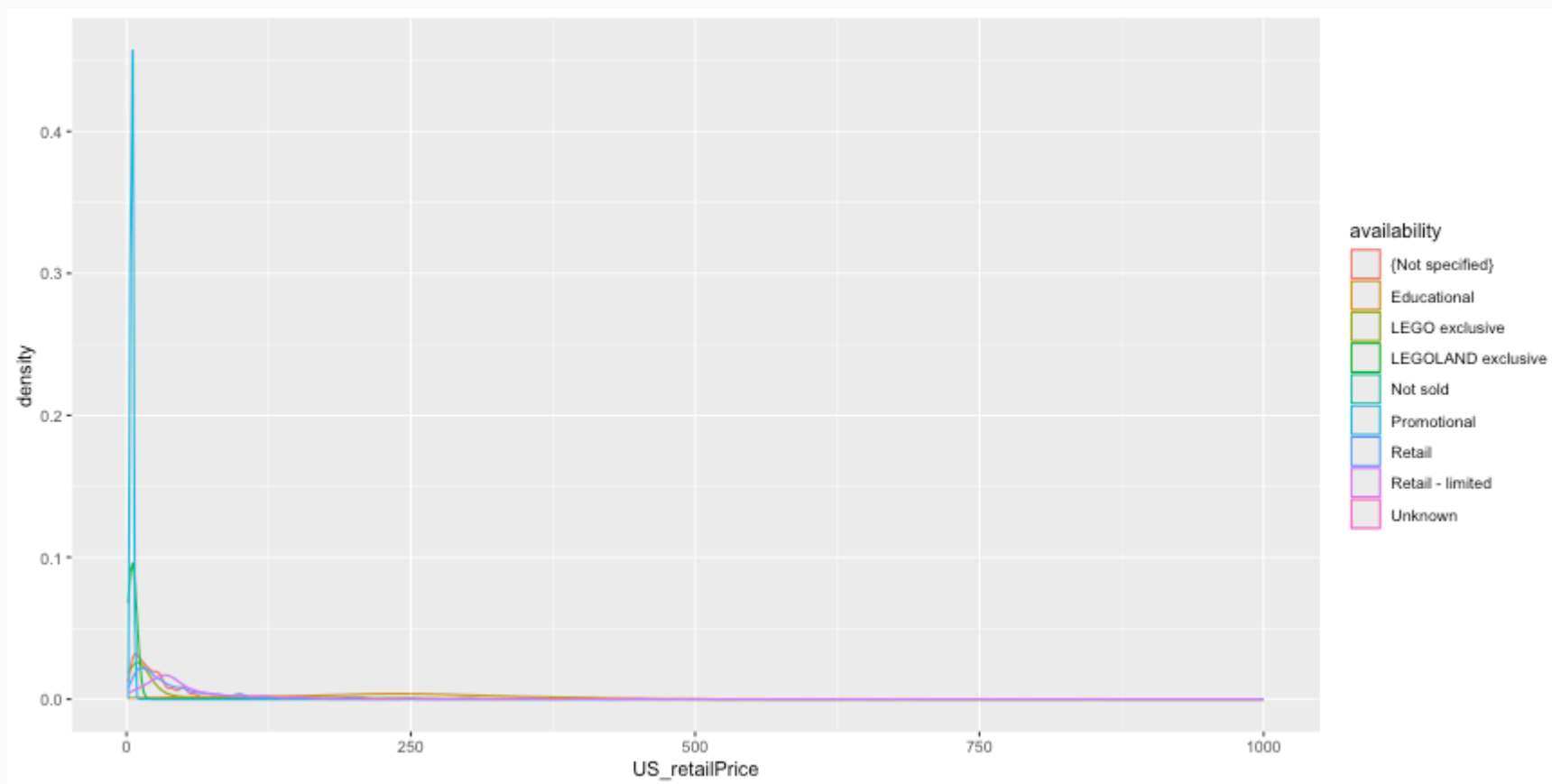
```
ggplot(legosets, aes(x = US_retailPrice)) + geom_histogram(binwidth = 25) + facet_wrap(~ availability)
```



# Density Plots



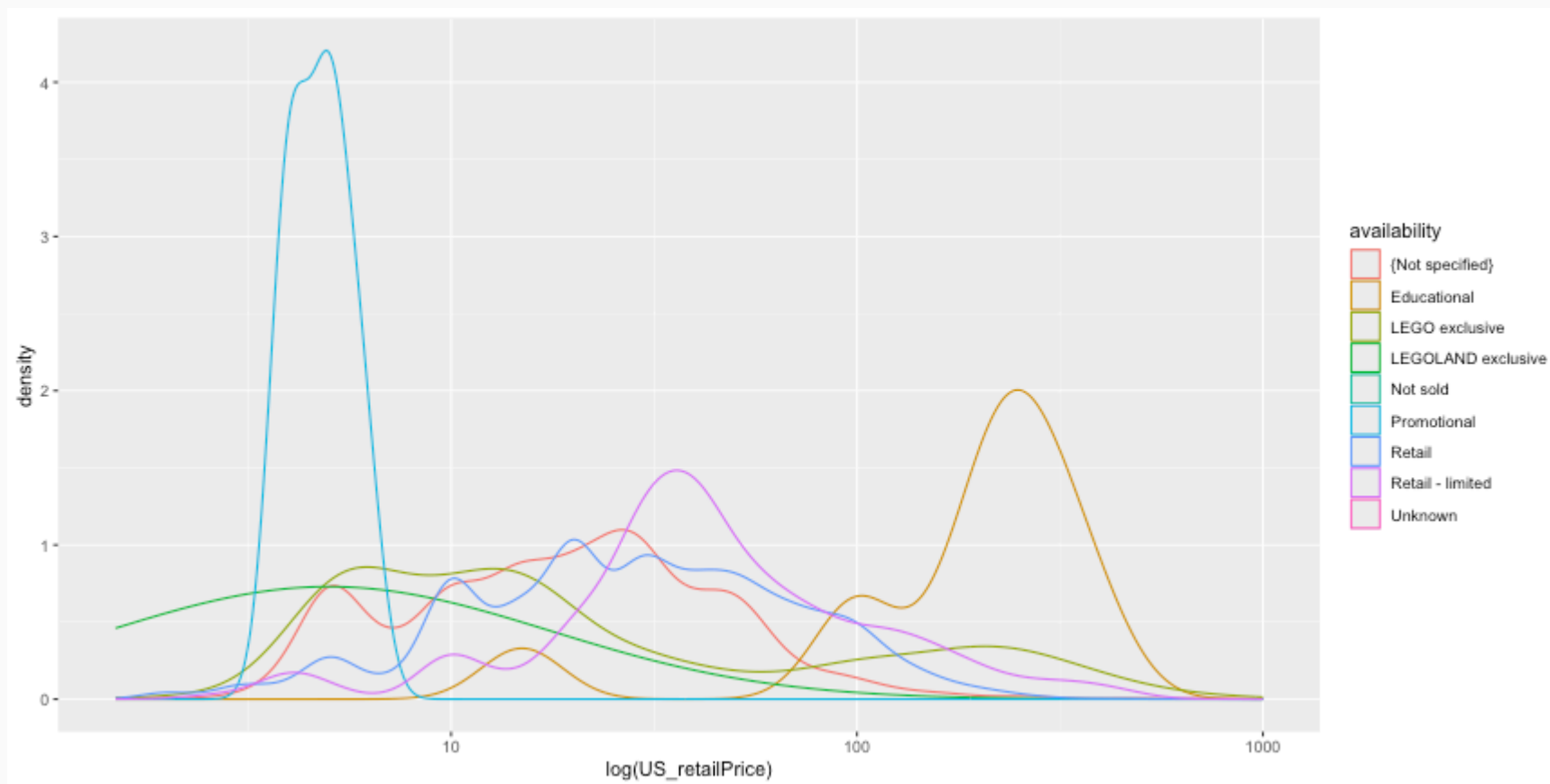
```
ggplot(legosets, aes(x = US_retailPrice, color = availability)) + geom_density()
```



# Density Plots (cont.)



```
ggplot(legosets, aes(x = US_retailPrice, color = availability)) + geom_density() + scale_x_log10() + xlab('log(US_ret
```



## ggplot2 aesthetics cheat sheet

Use this table to find the right aesthetics for your geoms:

Aesthetics that usually must be mapped to the data: use inside `aes()`

Aesthetics that can be mapped to the data: use in or outside `aes()`

Aesthetics that cannot be mapped to the data: use outside `aes()`

e.g., `ggplot(mpg, aes(x = class, y = displ)) + geom_col(aes(fill = class), width = .9)`

|                           | color | linetype | fill | y    | xmax | ymax | yend   | shape  | width  | angle  | hjust | label  | fontface   |                           |
|---------------------------|-------|----------|------|------|------|------|--------|--------|--------|--------|-------|--------|------------|---------------------------|
| group                     | size  | alpha    | x    | xmin | ymin | xend | weight | stroke | height | radius | vjust | family | lineheight |                           |
| area                      |       |          |      |      |      |      |        |        |        |        |       |        |            | area                      |
| bar (vertical)            |       |          |      |      |      |      |        |        |        |        |       |        |            | bar (vertical)            |
| bar (horizontal)          |       |          |      |      |      |      |        |        |        |        |       |        |            | bar (horizontal)          |
| bin2d                     |       |          |      |      |      |      |        |        |        |        |       |        |            | bin2d                     |
| boxplot                   |       |          |      |      |      |      |        |        |        |        |       |        |            | boxplot                   |
| col                       |       |          |      |      |      |      |        |        |        |        |       |        |            | col                       |
| contour                   |       |          |      |      |      |      |        |        |        |        |       |        |            | contour                   |
| contour_filled            |       |          |      |      |      |      |        |        |        |        |       |        |            | contour_filled            |
| count                     |       |          |      |      |      |      |        |        |        |        |       |        |            | count                     |
| crossbar (vertical)       |       |          |      |      |      |      |        |        |        |        |       |        |            | crossbar (vertical)       |
| crossbar (horizontal)     |       |          |      |      |      |      |        |        |        |        |       |        |            | crossbar (horizontal)     |
| curve                     |       |          |      |      |      |      |        |        |        |        |       |        |            | curve                     |
| density                   |       |          |      |      |      |      |        |        |        |        |       |        |            | density                   |
| density_2d                |       |          |      |      |      |      |        |        |        |        |       |        |            | density_2d                |
| dotplot                   |       |          |      |      |      |      |        |        |        |        |       |        |            | dotplot                   |
| errorbar                  |       |          |      |      |      |      |        |        |        |        |       |        |            | errorbar                  |
| errorbarh                 |       |          |      |      |      |      |        |        |        |        |       |        |            | errorbarh                 |
| freqpoly                  |       |          |      |      |      |      |        |        |        |        |       |        |            | freqpoly                  |
| hex                       |       |          |      |      |      |      |        |        |        |        |       |        |            | hex                       |
| histogram (on x-axis)     |       |          |      |      |      |      |        |        |        |        |       |        |            | histogram (on x-axis)     |
| histogram (on y-axis)     |       |          |      |      |      |      |        |        |        |        |       |        |            | histogram (on y-axis)     |
| jitter                    |       |          |      |      |      |      |        |        |        |        |       |        |            | jitter                    |
| label                     |       |          |      |      |      |      |        |        |        |        |       |        |            | label                     |
| line                      |       |          |      |      |      |      |        |        |        |        |       |        |            | line                      |
| linrange (vertical)       |       |          |      |      |      |      |        |        |        |        |       |        |            | linrange (vertical)       |
| linrange (horizontal)     |       |          |      |      |      |      |        |        |        |        |       |        |            | linrange (horizontal)     |
| map                       |       |          |      |      |      |      |        |        |        |        |       |        |            | map                       |
| path                      |       |          |      |      |      |      |        |        |        |        |       |        |            | path                      |
| point                     |       |          |      |      |      |      |        |        |        |        |       |        |            | point                     |
| pointrange (vertical)     |       |          |      |      |      |      |        |        |        |        |       |        |            | pointrange (vertical)     |
| pointrange (horizontal)   |       |          |      |      |      |      |        |        |        |        |       |        |            | pointrange (horizontal)   |
| polygon                   |       |          |      |      |      |      |        |        |        |        |       |        |            | polygon                   |
| quantile                  |       |          |      |      |      |      |        |        |        |        |       |        |            | quantile                  |
| raster                    |       |          |      |      |      |      |        |        |        |        |       |        |            | raster                    |
| rect                      |       |          |      |      |      |      |        |        |        |        |       |        |            | rect                      |
| ribbon (variation y-axis) |       |          |      |      |      |      |        |        |        |        |       |        |            | ribbon (variation y-axis) |
| ribbon (variation x-axis) |       |          |      |      |      |      |        |        |        |        |       |        |            | ribbon (variation x-axis) |
| rug                       |       |          |      |      |      |      |        |        |        |        |       |        |            | rug                       |
| segment                   |       |          |      |      |      |      |        |        |        |        |       |        |            | segment                   |
| smooth                    |       |          |      |      |      |      |        |        |        |        |       |        |            | smooth                    |
| spoke                     |       |          |      |      |      |      |        |        |        |        |       |        |            | spoke                     |
| step                      |       |          |      |      |      |      |        |        |        |        |       |        |            | step                      |
| text                      |       |          |      |      |      |      |        |        |        |        |       |        |            | text                      |
| tile                      |       |          |      |      |      |      |        |        |        |        |       |        |            | tile                      |
| violin                    |       |          |      |      |      |      |        |        |        |        |       |        |            | violin                    |

● usually must be inside `aes()` ■ can be inside `aes()` ◆ must be outside `aes()`

idea and design: Christian Burkhardt  
design advice: Ida Aarnio



Likert scales are a type of questionnaire where respondents are asked to rate items on scales usually ranging from four to seven levels (e.g. strongly disagree to strongly agree).

```
library(likert)
library(reshape2)
data(pisaitems)
items24 <- pisaitems[,substr(names(pisaitems), 1,5) == 'ST24Q']
items24 <- items24 |> dplyr::rename(
  "I read only if I have to." = ST24Q01,
  "Reading is one of my favorite hobbies." = ST24Q02,
  "I like talking about books with other people." = ST24Q03,
  "I find it hard to finish books." = ST24Q04,
  "I feel happy if I receive a book as a present." = ST24Q05,
  "For me, reading is a waste of time." = ST24Q06,
  "I enjoy going to a bookstore or a library." = ST24Q07,
  "I read only to get information that I need." = ST24Q08,
  "I cannot sit still and read for more than a few minutes." = ST24Q09,
  "I like to express my opinions about books I have read." = ST24Q10,
  "I like to exchange books with my friends." = ST24Q11)
```





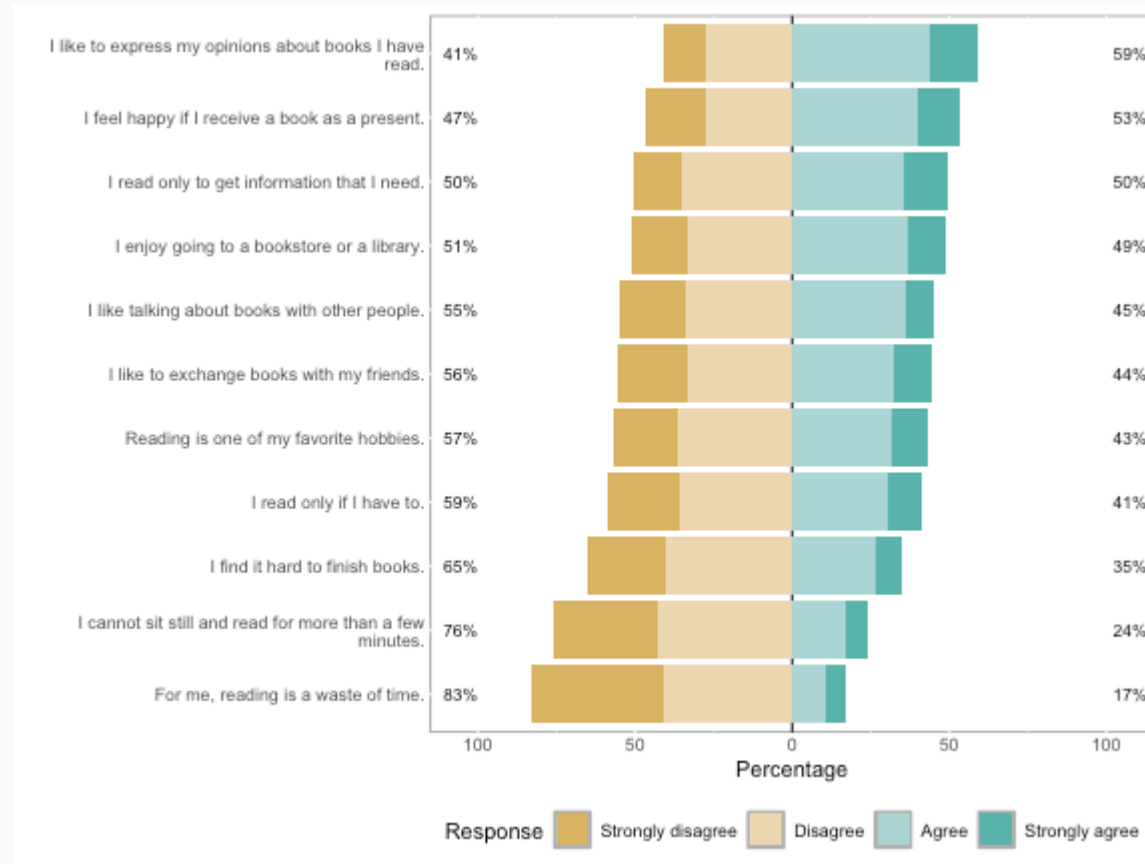
```
l24 <- likert(items24)
summary(l24)
```

| ##    | Item                                                     | low      | neutral | high     | mean     | sd        |
|-------|----------------------------------------------------------|----------|---------|----------|----------|-----------|
| ## 10 | I like to express my opinions about books I have read.   | 41.07516 | 0       | 58.92484 | 2.604913 | 0.9009968 |
| ## 5  | I feel happy if I receive a book as a present.           | 46.93475 | 0       | 53.06525 | 2.466751 | 0.9446590 |
| ## 8  | I read only to get information that I need.              | 50.39874 | 0       | 49.60126 | 2.484616 | 0.9089688 |
| ## 7  | I enjoy going to a bookstore or a library.               | 51.21231 | 0       | 48.78769 | 2.428508 | 0.9164136 |
| ## 3  | I like talking about books with other people.            | 54.99129 | 0       | 45.00871 | 2.328049 | 0.9090326 |
| ## 11 | I like to exchange books with my friends.                | 55.54115 | 0       | 44.45885 | 2.343193 | 0.9609234 |
| ## 2  | Reading is one of my favorite hobbies.                   | 56.64470 | 0       | 43.35530 | 2.344530 | 0.9277495 |
| ## 1  | I read only if I have to.                                | 58.72868 | 0       | 41.27132 | 2.291811 | 0.9369023 |
| ## 4  | I find it hard to finish books.                          | 65.35125 | 0       | 34.64875 | 2.178299 | 0.8991628 |
| ## 9  | I cannot sit still and read for more than a few minutes. | 76.24524 | 0       | 23.75476 | 1.974736 | 0.8793028 |
| ## 6  | For me, reading is a waste of time.                      | 82.88729 | 0       | 17.11271 | 1.810093 | 0.8611554 |

# likert Plots

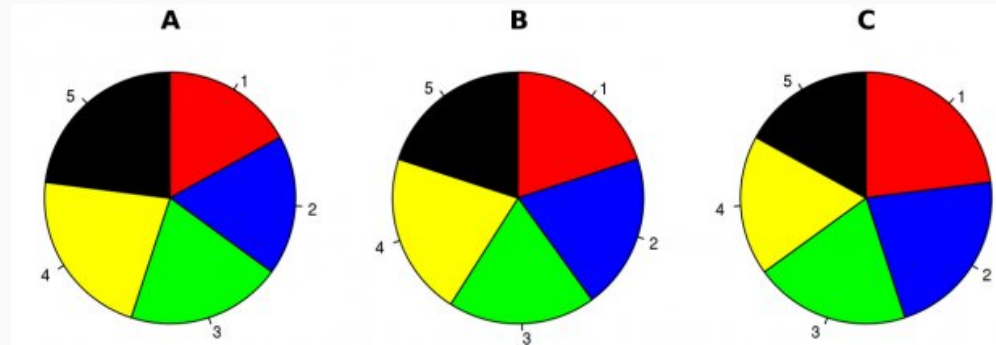


```
plot(l24)
```



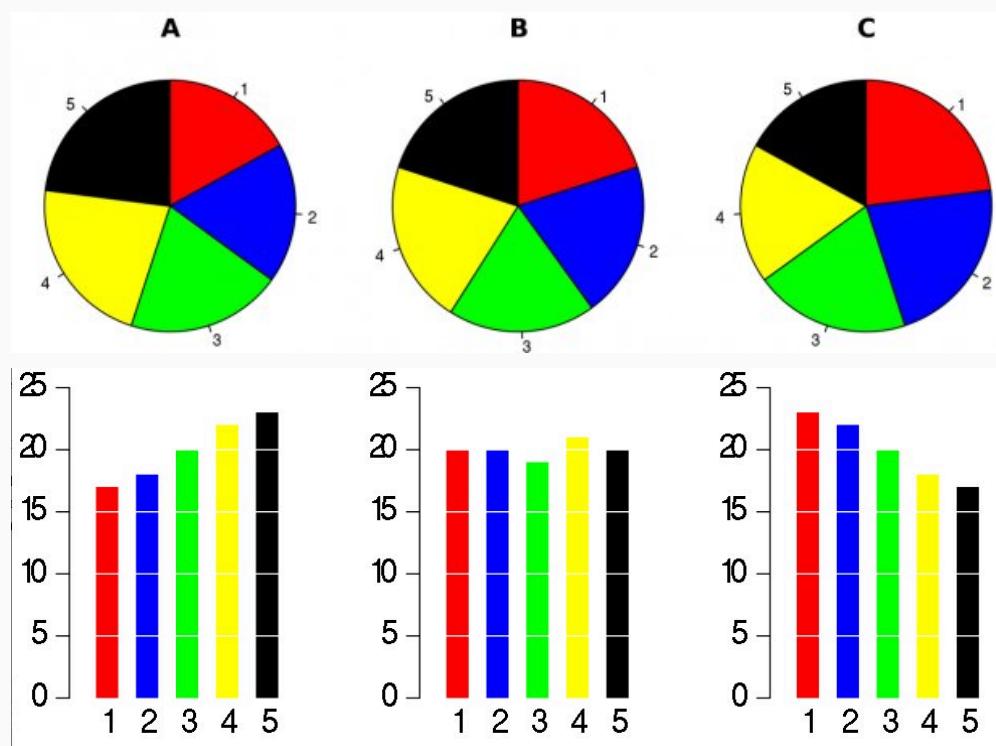
# Pie Charts

There is only one pie chart in *OpenIntro Statistics* (Diez, Barr, & Çetinkaya-Rundel, 2015, p. 48). Consider the following three pie charts that represent the preference of five different colors. Is there a difference between the three pie charts? This is probably a difficult to answer.



# Pie Charts

There is only one pie chart in *OpenIntro Statistics* (Diez, Barr, & Çetinkaya-Rundel, 2015, p. 48). Consider the following three pie charts that represent the preference of five different colors. Is there a difference between the three pie charts? This is probably a difficult to answer.



"There is no data that can be displayed in a pie chart that cannot better be displayed in some other type of chart"

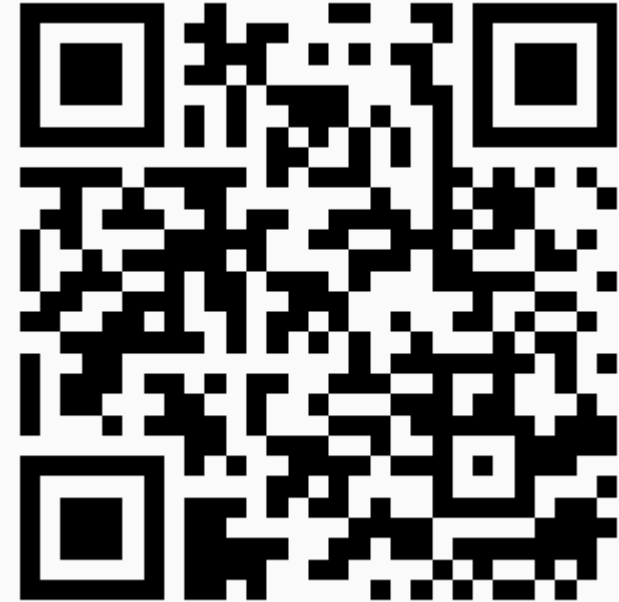
John Tukey

# Additional Resources

- ggplot2 website: <https://ggplot2.tidyverse.org>
- R for Data Science book: <https://r4ds.had.co.nz/data-visualisation.html>
- R Graphics Cookbook: <https://r-graphics.org>
- Data visualization cheat sheet: <https://github.com/rstudio/cheatsheets/raw/master/data-visualization-2.1.pdf>

# One Minute Paper

1. What was the most important thing you learned during this class?
2. What important question remains unanswered for you?



<https://forms.gle/hWUktVZ4Fyiia3xy6>